

Andrei Katsevich

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Education

Princeton University

GRADUATE STUDENT WORKING ON HIGH ENERGY THEORY, PHYSICS DEPARTMENT

- Advisor: [Igor R. Klebanov](#) (Princeton University)

Princeton, NJ, USA

August 2023 - Present

Moscow Institute of Physics and Technology (MIPT)

BACHELOR'S DEGREE IN APPLIED MATHEMATICS AND PHYSICS, LANDAU PHYSTECH SCHOOL OF PHYSICS AND RESEARCH

EDUCATIONAL PROGRAM «QUANTUM FIELD THEORY, STRING THEORY AND MATHEMATICAL PHYSICS»

- Bachelor's thesis: [Perturbed minimal models of two-dimensional conformal field theory](#) (in Russian)
- Advisor: [Alexey Litvinov](#) (Landau Institute)
- GPA: 4.98/5.00 (9.77/10.00) – ranked №1 of the department

Moscow, Russia

August 2019 - July 2023

Work Experience

Princeton University, Physics Department

TEACHING ASSISTANT

- Was TA at laboratory sessions in PHY103. General Physics I, PHY104. General Physics II and PHY106. Advanced Physics (Electromagnetism). Did grading of home assignments in PHY105. Advanced Physics (Mechanics).

Chernogolovka, Russia

July 2022 - July 2023

L.D. Landau Institute for Theoretical Physics of Russian Academy of Sciences

LABORATORY ASSISTANT RESEARCHER

- Studied QFT, CFT and matrix models.

Chernogolovka, Russia

July 2022 - July 2023

State Educational Institution "Brest Lyceum №1 named after A.S. Pushkin"

PHYSICS TEACHER

- Prepared lyceum students for Belarusian Republican and International Physics Olympiads.

Brest, Belarus

July 2020, 2021, 2022

Publications

The two-flavor Schwinger model at 50: Solving Coleman's puzzles [arXiv:2605.08042](#)

GABRIEL CUOMO, ROSS DEMPSEY, ANDREI KATSEVICH, IGOR R. KLEBANOV, ILIA V. KOCHERGIN, SILVIU S. PUFU, BENJAMIN T.

SØGAARD

- Using various methods (bosonization, integrability, numerics), we solve three Coleman's puzzles from his seminal paper.

May 2026

Preprint

$\mathcal{PT}$ -symmetric Field Theories at Finite Temperature [arXiv:2604.08459](#)

OLEKSANDR DIATLYK, ANDREI KATSEVICH, FEDOR K. POPOV

- We study the thermal properties of $\mathcal{PT}$ -symmetric field theories, compare quantities on a $6 - \epsilon$ -dimensional cylinder with exact OPE data in $d = 2$ CFTs.
- We check recent conjectures about GL descriptions of some minimal models.

April 2026

Preprint

Towards a Quintic Ginzburg-Landau Description of the $(2, 7)$ Minimal Model

[arXiv:2510.19085](#)

ANDREI KATSEVICH, IGOR R. KLEBANOV, ZIMO SUN, GRIGORY TARNOPOLSKY

- We resolve long-standing puzzle in the field about Ginzburg-Landau description of $M(2, 7)$ minimal model.

Sphere free energy of scalar field theories with cubic interactions [arXiv:2412.14086](#)

SIMONE GIOMBI, ELIZABETH HIMWICH, ANDREI KATSEVICH, IGOR R. KLEBANOV, ZIMO SUN

- We study the sphere free energy in cubic $O(N)$ model using $6 - \epsilon$ expansion: checked the violation of F -theorem for RG flow between non-unitary theories and proposed Long Range Approach.

October 2025

[PRL 136 \(2026\) 11, 111602](#)

October 2024

[JHEP 12 \(2025\) 133](#)

The spectrum of perturbed $(3, 10)$ minimal model [arXiv:2410.18069](#)

ANDREI KATSEVICH

- We study the spectrum of relevantly perturbed non-unitary minimal model $M(3, 10)$.
- Significant part of work was done in [MIPT Bachelor's thesis](#).

October 2024

Preprint

Ginzburg-Landau description of a class of non-unitary minimal models [arXiv:2410.11714](#)

ANDREI KATSEVICH, IGOR R. KLEBANOV, ZIMO SUN

- We study the GL description of a class of non-unitary minimal models $M(q, 3q \pm 1)$ with odd q .

October 2024

[JHEP 03 \(2025\) 170](#)

The eigenvalue spectrum of a large real antisymmetric random matrix with non-zero mean [arXiv:2309.01833](#)

September 2023

ANDREI KATSEVICH, PAVEL MESHCHERIAKOV

Preprint

- Using the replica method, we study the spectrum of a random antisymmetric real matrix with nonzero mean.

Talks at Conferences and Schools

Thermal Seminars

DESY

Zoom

June 16, 2026

- Future talk: $\mathcal{PT}$ -symmetric Field Theories at Finite Temperature.

Workshop on Confinement and QCD Strings

SCGP, STONY BROOK

Stony Brook

November 10-12, 2025

- Talk: [Ginzburg-Landau description of non-unitary minimal models](#).

Workshop on Non-perturbative Quantum Field Theory and Confinement

CARNEGIE MELLON UNIVERSITY

Pittsburgh

May 28-31, 2025

- Talk: [Comments on Ginzburg-Landau description of non-unitary minimal models](#).

4-th Spring Student School in Mathematics and Physics

SKOLTECH

Moscow

May 2023

- Gave two lectures on [S-matrix](#) and [perturbed Ising model](#).

Winter school ITMP MSU 2023

ORGANIZERS: [A. LITVINOV](#) (LANDAU INST.), [D. BYKOV](#) (MSU), [S. MIRONOV](#) (Moscow, ITEP), [D. PONOMAREV](#) (ITMP MSU)

Moscow

January-February 2023

- Gave a lecture on [perturbed minimal models of two-dimensional conformal field theory](#).

Participation at Conferences and Schools

2025 Simons Collaboration on Confinement and QCD Strings Annual Meeting

SIMONS FOUNDATION

New York

November 13-14, 2025

Prospects in Theoretical Physics (PiTP) 2025. Gravitational Waves from Theory to Observation

IAS

Princeton

July 2025

KITP conference Lattice and Continuum Approaches to Strongly Coupled QFT

UCSB, KITP

Santa Barbara

March 10-13, 2025

PCTS Confinement and QCD Strings Workshop

PRINCETON UNIVERSITY

Princeton

November 10-12, 2024

2024 Simons Collaboration on Confinement and QCD Strings Annual Meeting

SIMONS FOUNDATION

New York

November 7-8, 2024

Prospects in Theoretical Physics (PiTP) 2024. Quantum Matter Summer School

IAS

Princeton

July 2024

Simons Collaboration on Confinement and QCD Strings Annual Meeting

SIMONS FOUNDATION

New York

November 2-3, 2023

65th All-Russian Scientific Conference of MIPT

MIPT

Moscow

April 2023

- Report subject: [Truncated Conformal Space Approach for non-unitary minimal models](#) (in Russian).

Theoretical Physics Summer Practicum 2022

ORGANIZERS: L. LEVITOV (MIT), A. BOYARSKY (EPFL), L. SPODYNEIKO (CALTECH, MIT), G. TARNOPOLSKY (CARNEGIE MELLON UNIVERSITY), A. ZHIBOEDOV (CERN) AND OTHERS WITH SUPPORT OF KNU AND KHARKIV UNIVERSITY

Remote

August 2022

- Participated in the project of [Grigory Tarnopolsky](#) "Introduction to Random Matrix Theories in Quantum Physics".

MAY 27, 2026

Theoretical Physics Summer Practicum 2021

Remote

ORGANIZERS: L. LEVITOV (MIT), D. ABANIN (PRINCETON UNIVERSITY), N. NEKRASOV (SIMONS CENTER FOR GEOMETRY AND PHYSICS), A. SADOFYEV (SANTIAGO DE COMPOSTELA U., IGFAE) AND OTHERS WITH SUPPORT OF SKOLTECH, MIPT AND HSE

August 2021

- Was selected for the Practicum, being in the 1-st place in the competition for solving problems from the initial sections of physics.
- Participated in the project of [Andrey Sadofyev](#) "Hydrodynamics as an Effective Field Theory".

Skills

Physical background	General Physics, Analytical mechanics, Classical field theory, General theory of relativity, Cosmology, Quantum Mechanics, Quantum field theory, Conformal field theory, Statistical physics
Mathematical background	Mathematical analysis, Linear algebra, Differential equations, Complex analysis, Probability Theory, Lie groups and algebras and their representations, Integrable systems, Random matrix theory
Landau theoretical minimums	Mathematics I, Mechanics, Field theory, Mathematics II, Quantum mechanics, Quantum electrodynamics
Other	LaTeX (some of my solutions , mostly in Russian), Wolfram Mathematica, You-Tube channel

Honors

GRADUATE SCHOOL (PRINCETON UNIVERSITY)

2023-2024 **The Elliott Robinson Little '25 Student Aid Fund**, Named Fellowship

Princeton, NJ, USA

UNDERGRADUATE SCHOOL (MIPT)

2022-2023 **Prize-winner (2023), Absolute Winner (2022)**, [MIPT Olympiad in Theoretical Physics](#)

Moscow, Russia

2022-2023 **Prize-winner (2023), Winner (2022)**, [All-Russian Student Olympiad "I'm a professional" in Physics](#)

Moscow, Russia

2022 **Prize-winner**, [All-Russian Student Olympiad "I'm a professional" in Mathematics](#)

Moscow, Russia

2022-2021 **Winner (2022), Prize-winner (2021)**, [MIPT Traditional Student Olympiad in Physics](#)

Moscow, Russia

HIGH SCHOOL

2019 **Bronze medal**, [50th International Physics Olympiad \(IPhO\)](#)

Tel Aviv, Israel

2019 **1-st Degree Diploma (Absolute Winner)**, International Mathematical Olympiad of the Borderland for students of grades 9-11 (Brest-Bialystok-Vilnius-Smolensk-Odessa)

Brest, Belarus

2019-2017 **3-rd (2017), 1-st (2018), 2-nd (2019) Degree Diploma**, Belarusian Republican Physics Olympiad (BelPhO)

Belarus

2018 **2-nd Place Diploma**, Physics Olympiad on International Belarusian-Syrian session for gifted children

NDC "Zubrenok",
Belarus

Extracurricular Activity

[Princeton Slavic Club](#)

Princeton, USA

4TH PRESIDENT

October 2024 - present

- Cultural/diversity organization for graduate students from Belarus, Russia, Ukraine and other post-soviet countries.
- Provided cultural events (New Year's Eve, Maslenitsa, Student Day) and socials (Movie and Board Game Nights).

MIPT Mentorship

Moscow, Russia

MENTOR

September 2022 - May 2023

- Fall 2022 mentor project: "Introduction to Random Matrix Theory".
- Spring 2023 lecture course: [Introduction to Statistical Physics: from Ising Model to Renormalization Group](#) (in Russian)

MIPT Visiting Olympiad at the SEI "Brest Lyceum №1 named after A.S. Pushkin"

Brest, Belarus

MAIN ORGANIZER

January 2020, 2021, 2022, 2023

- Four times ([2023](#), [2022](#), [2021](#), [2020](#)) held the Visiting Olympiad in Brest, which was attended by about 60 schoolchildren every year.